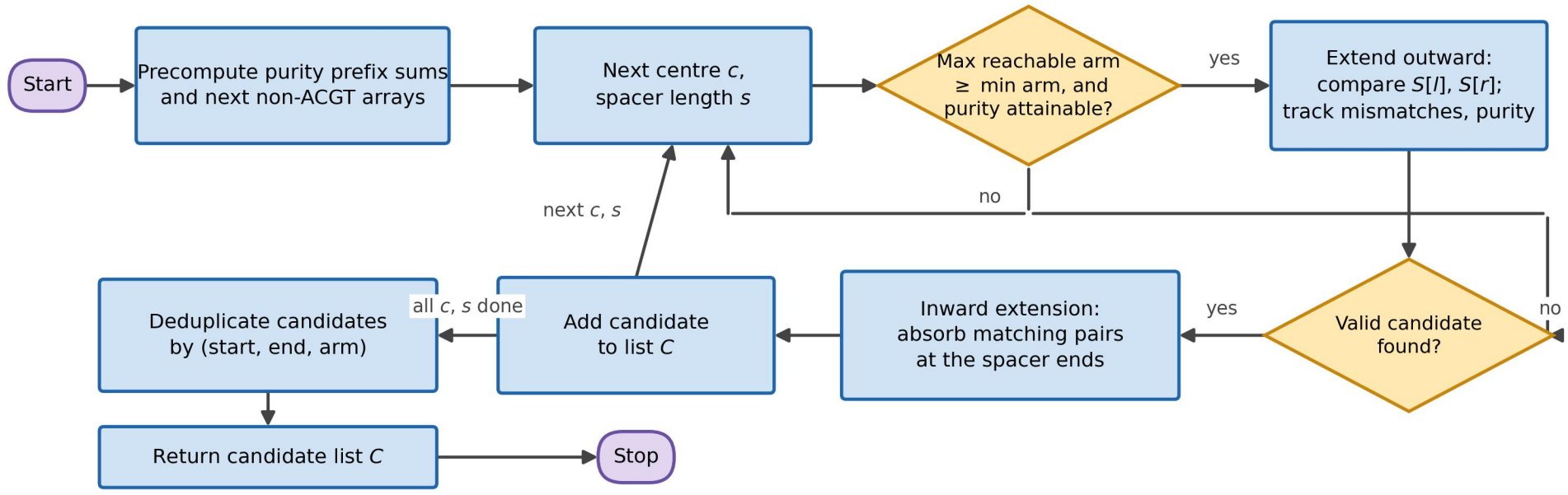
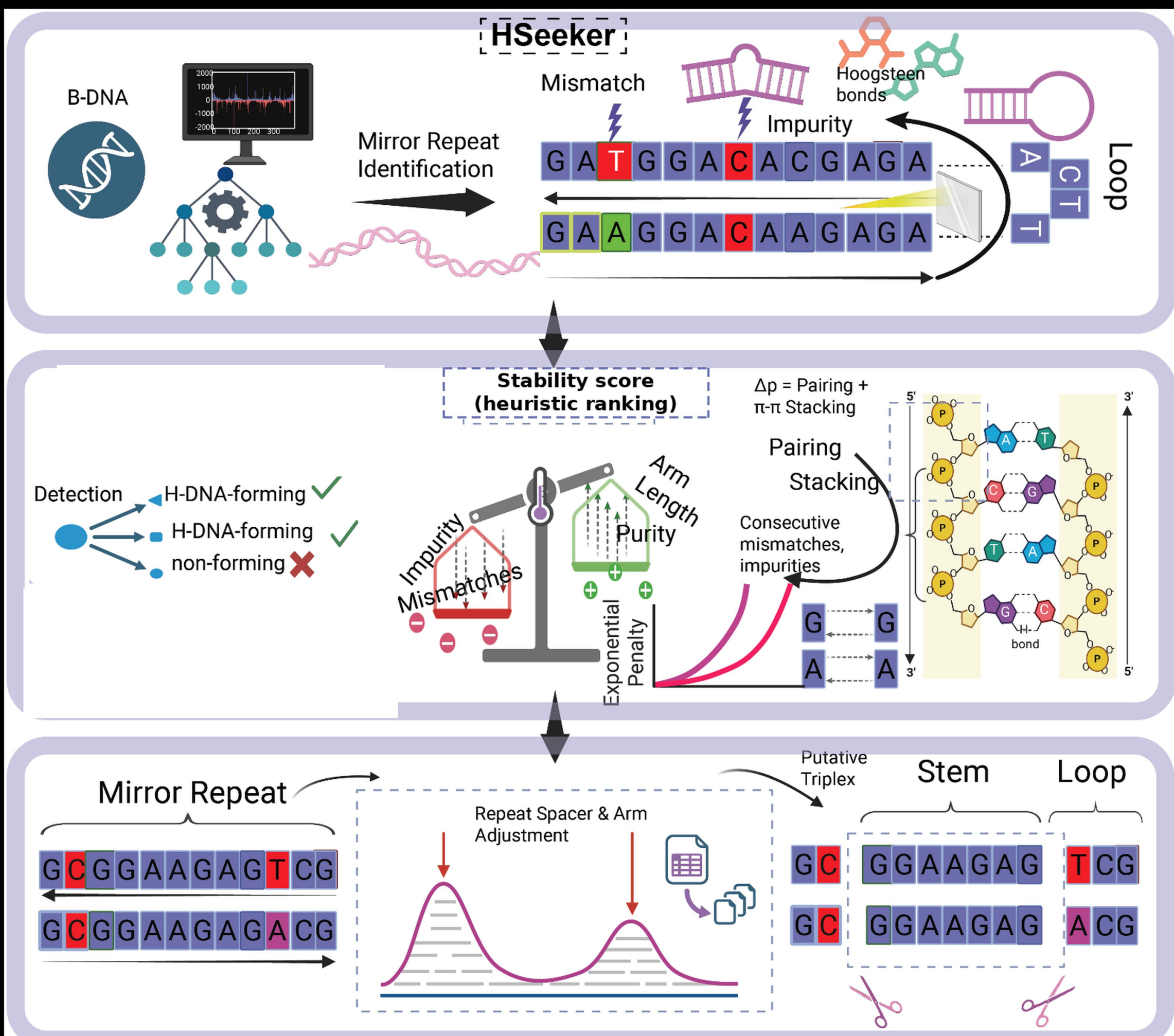
A


Start
Precompute purity prefix sums and next non-ACGT arrays
Next centre c, spacer length s
Max reachable arm ≥ min arm, and purity attainable?
yes
Extend outward: compare S[l], S[r]; track mismatches, purity
no
next c, s
Valid candidate found?
yes
no
Inward extension: absorb matching pairs at the spacer ends
Add candidate to list C
all c, s done
Deduplicate candidates by (start, end, arm)
Return candidate list C
Stop


HSeeker
B-DNA
Mismatch
Hoogsteen bonds
Impurity
Mirror Repeat Identification
G A T G G A C A C G A G A
G A A G G A C A A G A G A
A C T
T
Loop
Detection
H-DNA-forming
H-DNA-forming
non-forming
Stability score (heuristic ranking)
Impurity
Mismatches
Arm Length
Purity
Exponential Penalty
Δp = Pairing + π-π Stacking
Pairing
Stacking
Consecutive mismatches, impurities
G G
A A
5'
3'
H-bond
Mirror Repeat
G C G G A A G A G T C G
G C G G A A G A G A C G
Repeat Spacer & Arm Adjustment
Putative Triplex
Stem
Loop
G C G G A A G A G T C G
G C G G A A G A G A C G